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Detecting Careless Responding

A Permutation-Based Coherence Index and a Random
Forest Ensemble

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Chapter 1

Careless Responding Detection: The Field, Current Practices, and Software

1.1 The Problem of Careless Responding

Self-report questionnaires rest on a tacit assumption: that respondents read each item, retrieve the relevant information, form a judgment, and map it onto the response scale. When this process breaks down—when respondents answer without attending to item content—the resulting data carry the surface form of valid measurement but none of its substance. This phenomenon, variously labeled *careless responding*, *insufficient effort responding* (IER), *inattentive responding*, or *random responding*, has become one of the most actively studied threats to the validity of survey research (Curran, 2016; Meade & Craig, 2012; Ward & Meade, 2023).

The terminological proliferation reflects subtly different emphases. Huang et al. (2012) introduced *insufficient effort responding* to foreground the motivational mechanism: respondents who are unwilling to expend the cognitive effort the task demands. Krosnick (1991) had earlier framed the same behavior as *satisficing*—respondents who, rather than *optimizing* their answer through the full cognitive sequence, settle for a response that is merely good enough, or in the limiting case of *strong* satisficing, select responses with little or no reference to item content. *Careless responding* is the broader umbrella term now favored in methodological reviews (Ward & Meade, 2023); it encompasses both the low-effort respondent and the respondent whose attention lapses intermittently over a long instrument. Throughout this thesis I use *careless responding* as the cover term and reserve more specific labels for particular response patterns.

Whatever the label, the empirical consequences are well documented and consistently damaging. Careless responding inflates measurement error and depresses scale reliability (Huang et al., 2012). Because careless responses are, to a first approximation, unrelated to the constructs a questionnaire targets, they attenuate true relationships between variables; yet they can also *create* spurious relationships, most insidiously when careless respondents endorse the midpoint or an extreme of every scale and thereby manufacture artifactual covariance across otherwise unrelated measures (Huang et al., 2015). Maniaci and Rogge (2014) demonstrated across several studies that even modest proportions of inattentive participants can meaningfully distort effect sizes, suppress genuine effects, and reduce statistical power. Kam and Meyer (2015) showed that careless and acquiescent responding can distort the very factor structure recovered from a set of items, splitting a unidimensional construct into spurious method factors. In clinical and applied settings the stakes are higher still: Conijn et al. (2019) found that careless responding in routine outcome monitoring biased symptom estimates in ways that could affect treatment decisions.

Estimates of prevalence vary with population, mode of administration, and detection method, but they are rarely negligible. Reviews place the typical incidence somewhere between roughly 5% and 15% of respondents in well-controlled samples, with substantially higher rates in unsupervised online panels and crowdsourced labor markets (Curran, 2016; Meade & Craig, 2012; Ward & Meade, 2023). Bowling et al. (2016) added an important nuance: carelessness is not purely situational noise but is partly a stable individual difference, correlated with low conscientiousness and low agreeableness, meaning that the same respondents tend to provide low-effort data across studies. The practical upshot, emphasized in every recent treatment of the topic, is that careless responding cannot be assumed away: it must be actively prevented where possible and detected and addressed where not (DeSimone & Harms, 2018; DeSimone et al., 2015; Ward & Meade, 2023).

1.2 A Taxonomy of Causes and Response Patterns

Effective detection requires a clear picture of *what careless responding looks like* in the data, because different careless behaviors leave different statistical fingerprints. The literature converges on a useful organizing distinction between *consistent* and *inconsistent* careless responding (Curran, 2016; Ward & Meade, 2023).

Consistent careless patterns are those in which the respondent produces a regular, low-information sequence. The clearest example is *straightlining* (also called *non-differentiation*): selecting the same response option for a long run of consecutive items, often the scale midpoint or an extreme. A closely related notion is the *longstring*—the maximum length of consecutive

identical responses, which Johnson (2005) established as a simple but effective marker of inattentive protocols. *Acquiescent* responding (uniform agreement regardless of item keying) is another consistent pattern, particularly damaging because it survives naive reliability checks while corrupting the covariance structure (Kam & Meyer, 2015). Consistent patterns are, paradoxically, both easy and hard to detect: easy because their regularity is conspicuous to a longstring counter, hard because a respondent who endorses the midpoint of every item can look indistinguishable from a genuinely neutral one.

A methodological caveat sharpens this picture and bears directly on the design of the detector developed in this thesis. Best practice in questionnaire construction is to *randomize the order in which items are presented* to each respondent, which controls for order and context effects and breaks up contiguous blocks of same-construct items (Ward & Meade, 2023). This otherwise desirable practice substantially undermines the longstring index. A careless respondent who repeatedly selects the same option no longer produces a long *contiguous* run of identical responses once the data are returned to their canonical, construct-grouped order for analysis: the repeated answers are scattered across non-adjacent positions, and reverse-keyed items, after recoding, break whatever run remains. Under randomized administration the longstring counter therefore loses much of its sensitivity, and even straightlining—the very pattern it was designed to catch—can slip past it. This is one reason the simulation studies in later chapters model longstrings as *non-contiguous*, scattered across the response vector rather than as neat adjacent blocks: a deliberately conservative choice that reflects modern randomized administration and that motivates detectors whose signal does not depend on the contiguity or ordering of responses.

Inconsistent careless patterns, by contrast, involve responses that vary across items but bear no coherent relationship to item content. The prototype is *purely random* responding, in which answers are effectively drawn at random from the response scale. Such protocols display normal-looking variance and no long runs, so they slip past longstring and variability checks; what betrays them is the *absence of the correlational structure* that attentive respondents exhibit—answers to items measuring the same latent trait, which should covary, instead scatter (Curran, 2016; Meade & Craig, 2012). *Mixed* patterns, in which a respondent attends to part of the instrument and disengages for the rest (for example through mid-survey *fatigue*), combine features of both families and are correspondingly difficult to catch with any single index (Ward & Meade, 2023).

This taxonomy matters for a reason that runs through the entire thesis: the indices in widest use were largely designed for, and excel at, consistent careless patterns, whereas inconsistent patterns—random and mixed responding—are precisely where most methods are weakest. The cross-factor coherence approach developed in later chapters is motivated by this gap.

1.3 Prevention by Design

Because no detection method is perfect, the first line of defense is to design instruments and data-collection procedures that discourage careless responding and that embed signals of inattention directly into the questionnaire (DeSimone et al., 2015; Ward & Meade, 2023).

The most common device is the *attention check*, which takes several forms. *Instructed-response items* (IRIs) direct the respondent to select a specific option (“Please select *strongly agree* for this item”); a deviation flags inattention. *Bogus items* assert something patently true or false (“I was born on February 30th”); an implausible answer is diagnostic. *Instructional manipulation checks* (IMCs), introduced by Oppenheimer et al. (2009), embed a comprehension test within a seemingly ordinary item and have been shown to identify respondents who are satisficing rather than reading. A self-report single-item approach simply asks respondents, at the end, whether their data should be used—a surprisingly informative question when posed without penalty (Meade & Craig, 2012).

These instruments are valuable but limited, and the limitations are instructive. Curran and Hauser (2019) showed that attention-check items generate a non-trivial rate of *valid-but-incorrect* responses: attentive participants sometimes fail them (through misreading or genuine ambiguity), while determined careless respondents can pass an isolated check by chance or by briefly attending to a conspicuous item. A single check therefore has limited sensitivity and imperfect specificity. Attention checks also consume questionnaire space, can alter respondent behavior by signaling surveillance, and—being placed at fixed positions—catch inattention only at those points, missing the intermittent disengagement characteristic of fatigue. For time-stamped or computer-administered surveys, *response latency* offers a complementary design-based signal: implausibly fast page completions indicate that a respondent could not have read the items, a logic formalized by Wise and Kong (2005) in the *response time effort* measure and later embedded in model-based approaches (Ulitzsch et al., 2022). Response times, however, are unavailable in the many settings where only the final responses are recorded.

The recurring lesson is that prevention reduces but does not eliminate careless responding, and that the design-based signals it provides are partial. This motivates *post-hoc* detection from the response data themselves—methods that require no special items, no timing information, and no advance planning, and that can therefore be applied to data already collected.

1.4 Post-hoc Identification Methods

Post-hoc methods infer carelessness from the structure of the observed responses. They fall into several families, each exploiting a different statistical signature of inattention. The two most authoritative syntheses of this literature—Meade and Craig (2012) and Curran (2016)—organize the field around a core set of indices that I summarize here and collect, with their software implementations, in Table 1.1.

1.4.1 Consistency and Response-Pattern Indices

Longstring counts the maximum number of consecutive identical responses in a protocol. It is the canonical detector of straightlining and is trivial to compute, but it is blind to inconsistent careless responding and is biased by the number of items keyed in the same direction (Johnson, 2005; Meade & Craig, 2012).

The *intra-individual response variability* (IRV) is the standard deviation of a respondent’s answers across a set of items (Dunn et al., 2018). Very low IRV signals straightlining (little or no variation), while implausibly high IRV can indicate random responding across the full scale range. Because it condenses a protocol to a single dispersion statistic, IRV is cheap and intuitive, but it discards all information about *which* items were answered how.

Psychometric synonyms and antonyms (Meade & Craig, 2012) operationalize coherence at the item-pair level. One identifies pairs of items that are strongly positively correlated in the sample (synonyms) or strongly negatively correlated (antonyms), then computes, *within each respondent*, the correlation between their answers to the two members of each pair. An attentive respondent answers synonym pairs similarly and antonym pairs oppositely, producing a high within-person correlation; a careless respondent does not. This is a powerful idea because it targets inconsistent careless responding directly, and it is the conceptual ancestor of the cross-factor coherence method developed later in this thesis.

The *person-total correlation* correlates a respondent’s profile of answers with the sample’s mean response to each item; respondents who answer against the grain of the typical pattern receive low values. *Even-odd consistency* splits a scale into even- and odd-numbered items, computes two subscale scores, and assesses their agreement; *resampled individual reliability* generalizes this by repeatedly splitting and correlating. These consistency indices trace back to classical work on individual-protocol validity (Jackson, 1976; Johnson, 2005).

1.4.2 Multivariate Outlier Detection

A complementary strategy treats careless protocols as multivariate *outliers*. The *Mahalanobis distance* (Mahalanobis, 1936) measures how far a respondent’s response vector lies from the sample centroid, scaled by the inverse covariance matrix, and is the most widely used outlier index in this literature (Curran, 2016; Meade & Craig, 2012). Under multivariate normality the squared distance follows a chi-square distribution with degrees of freedom equal to the number of items, which suggests a principled cutoff. In practice, Likert data violate the normality assumption badly, so the chi-square reference distribution is only approximate, and—as the simulation chapters of this thesis document in detail—the nominal cutoff can be far too conservative for questionnaire data, flagging almost no one. Mahalanobis distance also detects *any* multivariate outlier, careless or not, including genuinely unusual but attentive respondents.

1.4.3 Person-Fit and Item-Response-Theory Approaches

A distinct tradition grew out of item response theory (IRT), where *person-fit* statistics quantify how well an individual’s response pattern conforms to the fitted measurement model. The l_z statistic and the Rasch *infit/outfit* indices are the best known; Karabatsos (2003) compared thirty-six such statistics for detecting aberrant response patterns, and Meijer et al. (2016) provided a practical guide to applying them in clinical research. Person-fit methods rest on a strong measurement model, which is their strength (a principled probabilistic account of “aberrance”) and their weakness (sensitivity to model misspecification, and a need for the model to be estimable). More recently, *mixture-model* formulations treat the population as a blend of attentive and careless latent classes and estimate class membership directly; Ulitzsch et al. (2022) combined item responses with response times in a latent-response mixture model that simultaneously identifies and adjusts for careless and insufficient-effort responding.

1.4.4 Machine-Learning and Ensemble Approaches

The newest family abandons the idea of a single hand-crafted index in favor of *learning* a careless/attentive boundary from many indices at once. Schroeders et al. (2022) trained a stochastic gradient-boosting classifier on a battery of careless indicators and reported detection performance exceeding that of any constituent index, demonstrating that the indices carry *complementary* information that a flexible classifier can exploit. This ensemble logic is directly relevant to the present thesis, whose later chapters combine a novel coherence index with established detectors inside a Random Forest (Breiman, 2001). The turn toward machine learning has not gone uncontested. Alfons and Welz (2024) offer an open-science critique, warning that supervised approaches risk

overfitting to the idiosyncrasies of the simulated or instructed-careless data on which they are trained, that ground-truth labels for careless responding are inherently uncertain, and that reproducible benchmarks and shared data are largely absent from the field. Their related work on identifying the *onset* of careless responding within a protocol (Welz & Alfons, 2023) illustrates an alternative, model-based direction. These cautions frame the methodological commitments of this thesis—in particular the emphasis on validation across independent simulation draws and on deployment-realistic, label-free calibration.

1.5 Cutoffs, Thresholds, and Combining Indices

Computing an index is only half the task; the other half is deciding *where to draw the line*. Every method above yields a continuous score, and converting that score into a binary careless/attentive flag requires a threshold. This seemingly technical step is in fact one of the least settled and most consequential issues in the field.

Two broad strategies exist. The first uses a *theoretical* cutoff derived from a reference distribution—most prominently the chi-square cutoff for Mahalanobis distance—which has the appeal of requiring no tuning but, as noted, rests on distributional assumptions that questionnaire data routinely violate. The second uses an *empirical* cutoff: a fixed quantile of the observed scores, a visual “elbow” in the score distribution, or a value optimized against known labels. Empirical cutoffs adapt to the data but risk either circularity (when tuned on the same labels used to evaluate them) or arbitrariness (when chosen by eye). Ward and Meade (2023) note that the absence of consensus cutoffs is a genuine obstacle to cumulative science, because two analysts applying “the same” index to the same data can reach different conclusions about who is careless.

The comparative literature offers two robust findings that shape best practice. First, no single index dominates across conditions: Niessen et al. (2016) compared several methods on web-questionnaire data and concluded that the best choice depends on the type of careless responding present and on questionnaire characteristics, with different indices catching different careless respondents. Second—and following directly from the first—*combining* indices outperforms relying on any one of them (Curran, 2016; Schroeders et al., 2022). The recommended practice in recent reviews is therefore to compute multiple indices, examine their convergence, and flag respondents identified by several of them (DeSimone et al., 2015; Ward & Meade, 2023). The ensemble and threshold-calibration methods developed in this thesis are a formalization of exactly this advice: they replace ad hoc multi-index agreement rules with a trained classifier and a deployment-realistic operating point.

1.6 Software for Careless Responding Detection

The methods reviewed above are only as useful as the tools that implement them, and here the R ecosystem has become the de facto standard for careless responding detection. This section surveys the packages in current use; their indices and the constructs they target are summarized in Table 1.1.

The central package is **careless** (Yentes & Wilhelm, 2023), which provides a unified implementation of the core response-pattern indices: **longstring** (and its average-run variant), **irv**, **mahad** (Mahalanobis distance, with optional robust covariance estimation), **evenodd** (even-odd consistency), and **psychsyn/psychant** (psychometric synonyms and antonyms). Because it gathers the indices of Meade and Craig (2012) and Curran (2016) behind a consistent interface, **careless** is the package most often cited in applied work and the natural baseline against which new methods are compared.

Several other packages contribute pieces of the workflow. The **responsePatterns** package (Riháček & Gottfried, 2023) implements autocorrelation-based screening for repetitive response patterns (Gottfried et al., 2022), targeting the machine-like regularity that some careless protocols exhibit. For the multivariate-outlier branch, the **performance** package (Lüdtke et al., 2021) exposes **check_outliers**, which computes Mahalanobis (and several robust alternatives) within a tidy model-checking framework, while the venerable **psych** package (Revelle, 2024) supplies the factor-analytic and reliability machinery—exploratory factor analysis, parallel analysis, and scale-reliability estimation—on which several careless indices and most downstream psychometric analyses depend. Mixture-model approaches to careless detection draw on general-purpose finite-mixture engines such as **mclust** (Scrucca et al., 2016) and **mixtools** (Benaglia et al., 2009), which fit the latent attentive/careless class structure used by model-based methods. All of these run within base R (R Core Team, 2024), and supervised-ensemble approaches add a classifier layer—gradient boosting in Schroeders et al. (2022), or a Random Forest in the present work—on top of the index computations these packages provide.

What the ecosystem still lacks is as informative as what it offers. Alfons and Welz (2024) observe that, despite the maturity of the individual-index tooling, the field has no shared, labeled benchmark dataset against which competing detectors can be evaluated on equal terms, and that machine-learning detectors in particular are typically validated only on the simulated data their authors generate. There is, similarly, no standard, well-validated routine for *calibrating* an index threshold without access to ground-truth labels—the very problem the deployment-oriented chapters of this thesis address. Outside R, careless detection is comparatively underserved: commercial survey platforms offer little beyond fixed-position attention checks, and although general-purpose machine-learning libraries in other languages could in principle implement any of these methods, no established, careless-specific toolkit exists outside the R packages surveyed here.

Table 1.1: Principal careless-responding detection methods, the careless behavior each is best suited to detect, and their implementation in current R software.

Method	What it captures	Target pattern	R implementation
Longstring	Longest run of identical consecutive responses	Consistent (straightlining)	<code>careless::longstring</code>
IRV	SD of a respondent's answers across items	Consistent (low) / random (high)	<code>careless::irv</code>
Psychometric synonyms / antonyms	Within-person correlation over high- $ r $ item pairs	Inconsistent (random, mixed)	<code>careless::psychsyn</code> , <code>psychant</code>
Person-total correlation	Agreement of a profile with the sample mean profile	Inconsistent	<code>careless</code> / base R
Even-odd / re-sampled reliability	Internal consistency across item splits	Inconsistent	<code>careless::evenodd</code>
Mahalanobis D^2	Multivariate distance from the sample centroid	Multivariate outliers	<code>careless::mahad</code> ; <code>performance::check_outliers</code>
Autocorrelation screening	Machine-like repetitiveness of a protocol	Consistent (patterned)	<code>responsePatterns</code>
Person-fit (l_z , infit/outfit)	Departure from a fitted IRT model	Aberrant patterns	<code>mirt</code> / <code>PerFit</code>
Mixture / response-time models	Latent attentive vs. careless classes	All (model-based)	<code>mclust</code> , <code>mixtools</code>
Ensemble / machine learning	Learned combination of many indices	All	gradient boosting; Random Forest

1.7 Gaps in the Field and the Contribution of This Thesis

This review motivates the work that follows by making three gaps explicit.

First, the methods in widest use are strongest on *consistent* careless responding and weakest on the *inconsistent* patterns—random and mixed—that are arguably the hardest to prevent and the most damaging to covariance-based analyses. Longstring and IRV detect regularity; Mahalanobis distance detects outlyingness; but the cross-item *coherence* that distinguishes an attentive from a randomly responding participant is captured only partially, by the synonym/antonym indices, and only for short lists of hand-selected item pairs. A method that exploits the full correlational structure of a multi-construct questionnaire to detect inconsistent careless responding would fill a real gap. This is the motivation for the permutation-based coherence index,

ReReReRe, developed in Chapter 3.

Second, the field has converged on the principle that *combining* indices beats any single one (Curran, 2016; Niessen et al., 2016; Schroeders et al., 2022), but the dominant practice—flagging respondents identified by several indices under separately chosen cutoffs—remains informal. A trained classifier that learns the optimal combination of complementary detectors, and that is evaluated honestly across independent data, is the natural next step. This is the motivation for the Random Forest ensemble of Chapter 4.

Third, every detection method ultimately requires a *threshold*, and the field lacks a validated procedure for setting one without ground-truth labels that no analyst possesses in practice (Alfons & Welz, 2024; Ward & Meade, 2023). A calibration strategy that is both deployment-realistic and competitive with label-dependent oracles would make these methods usable by the applied researchers who need them. This is the motivation for the rate-aware calibration evaluated in Chapter 5.

The remaining chapters take up these gaps in turn. Chapter 2 reviews the statistical concepts on which the methods rest—correlation, exploratory factor analysis, the Random Forest classifier, and the evaluation metrics used throughout. Chapter 3 introduces *ReReReRe* and its underlying coherence principle. Chapter 4 develops the ensemble that combines *ReReReRe* with the established detectors surveyed above. Chapter 5 applies the ensemble under a deployment-realistic protocol and characterizes when, and how well, it works.

Chapter 2

Statistical Preliminaries

2.1 Orientation

The methods developed in the following chapters draw on a handful of statistical ideas that, while standard in statistics and machine learning, are not all part of the routine toolkit of a working psychologist. This chapter collects them in one place and explains them at the level of detail needed to follow the rest of the thesis. It is deliberately self-contained and tutorial in tone: a reader comfortable with these concepts may skim it, while a reader who last met some of them in an undergraduate methods course should find here enough to read Chapters 3 to 5 without consulting outside references.

The chapter falls into two halves. The first lays the *psychometric* foundations on which any discussion of questionnaire data rests—measurement scales, classical test theory, the latent-variable models of factor analysis and item response theory, and the twin criteria of reliability and validity—and shows how careless responding threatens each. The second introduces the *statistical and machine-learning* tools the detector itself is built from: permutation tests, the bootstrap, the Random Forest, cross-validation, and the metrics used to judge a classifier.

The selection is driven by what the method actually uses. The correlation coefficient (Section 2.2) is the raw material of the ReReReRe detector, which compares a respondent’s *individual-level* coherence with a baseline. Exploratory factor analysis (Section 2.6) supplies the latent-variable model that justifies why attentive responses to different items covary in the first place, and parallel analysis—the procedure ReReReRe uses to estimate the number of factors—is a factor-retention method. Permutation tests (Section 2.9) are the inferential engine behind ReReReRe’s per-respondent z-score, and the bootstrap (Section 2.10) provides the confidence intervals used to quantify the detector’s contribution. The Random Forest (Section 2.11) is the classifier that combines ReReReRe with the auxiliary detectors into an ensemble, cross-validation (Section 2.12) is how its performance is estimated

honestly, and the battery of evaluation metrics (Section 2.13)—above all the Matthews correlation coefficient—is how detection quality is reported throughout.

2.2 Correlation and Covariance

Two variables are *correlated* when knowing one tells us something about the other. The workhorse measure of linear association is the Pearson product-moment correlation coefficient (Pearson, 1896). For two variables X and Y measured on n units, it is defined as the covariance of X and Y divided by the product of their standard deviations,

$$r_{XY} = \frac{\text{cov}(X, Y)}{s_X s_Y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}. \quad (2.1)$$

The coefficient ranges from -1 (perfect negative linear association) through 0 (no linear association) to $+1$ (perfect positive linear association). The numerator, the *covariance*, is positive when above-average values of X tend to co-occur with above-average values of Y ; dividing by the standard deviations rescales this to the unit interval, making correlations comparable across variables measured on different scales.

Three properties of Equation 2.1 matter for what follows. First, r captures only *linear* association: two variables can be strongly related in a curved pattern yet have $r \approx 0$. Second, r is invariant to separate linear rescalings of X and Y : adding a constant to every value, or multiplying every value by a positive constant, leaves r unchanged. This invariance is exploited directly in Chapter 3, where it guarantees that dividing every item by its scale maximum—a rescaling used to handle mixed-format questionnaires—cannot change the correlation among items measured on a common scale. Third, the *sign* of r encodes the direction of the relationship, which is why reverse-keyed questionnaire items (worded so that agreement indicates *less* of the trait) correlate *negatively* with same-trait items keyed in the usual direction; handling that sign correctly is essential when aggregating correlations across item pairs.

A distinction that is central to this thesis, but easy to overlook, is between correlation computed *across respondents* and correlation computed *within a respondent*. The familiar use of Equation 2.1 treats each respondent as one unit and each pair of items as the two variables X and Y ; this *sample-level* correlation describes how two items relate in the population. One can instead fix a single respondent and treat their answers to a set of item pairs as the data, computing a correlation that describes whether *that person* answered coherently across pairs that the sample says should agree. This *individual-level* or *person-level* correlation is the quantity ReReReRe estimates and tests; the psychometric-synonym index of Chapter 1 is an early instance of the same idea (Meade & Craig, 2012).

2.3 Likert Scales and the Measurement Model

Most of the data this thesis concerns come from *Likert items* (Likert, 1932): statements to which respondents indicate agreement on an ordered set of categories, typically five or seven points from “strongly disagree” to “strongly agree.” Strictly, a Likert response is *ordinal*—the categories are ordered, but the psychological distance between “agree” and “strongly agree” need not equal that between “neutral” and “agree.” In practice, researchers routinely treat summed or averaged Likert responses as *interval* data and apply correlation- and mean-based statistics to them, a simplification that is generally defensible for scales with five or more categories that are reasonably symmetric.

The thesis follows this common practice, computing Pearson correlations on Likert responses, with two consequences worth flagging now. First, because the correlation in Equation 2.1 is sensitive to the numerical spacing of the response options, items administered on *different* Likert ranges (say, some on a 1–5 scale and others on a 1–7 scale) can contribute unequally to an individual-level correlation; Chapter 3 addresses this with a rescaling step. Second, a respondent who selects the same category for every item has zero variance, which makes the denominator of Equation 2.1 undefined for that person—a degenerate case that careless “straightlining” produces and that any individual-level method must handle explicitly.

2.4 Classical Test Theory

The oldest and still most widely used framework for thinking about test scores is *classical test theory* (CTT), axiomatised by Novick (1966) and given its definitive treatment by Lord and Novick (1968). CTT decomposes an observed score into two additive parts,

$$X = T + E, \quad (2.2)$$

where X is the score a respondent actually obtains, T is their *true score*—the score they would obtain on average over infinitely many independent administrations—and E is *random measurement error*. The theory’s defining assumptions are that error has expected value zero, is uncorrelated with the true score, and is uncorrelated across independent measurements; the observed-score variance then partitions cleanly into true and error variance, $\text{Var}(X) = \text{Var}(T) + \text{Var}(E)$.

Reliability is defined as the proportion of observed variance that is true variance,

$$\rho_{XX'} = \frac{\text{Var}(T)}{\text{Var}(X)} = 1 - \frac{\text{Var}(E)}{\text{Var}(X)},$$

a number between 0 (all noise) and 1 (no error); the related *standard error of measurement*, $\text{SEM} = s_X \sqrt{1 - \rho_{XX'}}$, expresses the same idea on the score

scale. CTT operates at the level of the *total score*; the latent-variable models of the next two sections instead model responses item by item.

Classical test theory gives the central problem of this thesis its sharpest formulation. A careless response carries little or no true-score information: in the limit of purely random responding, X is essentially all E and no T . Careless respondents therefore inflate $\text{Var}(E)$, which by definition lowers reliability, attenuates the correlations a scale was built to measure (Section 2.7), and biases any statistic computed on the contaminated data. Detecting careless responding is, in this language, the task of identifying the respondents whose observed scores are dominated by error rather than by true score—before they degrade the measurement for everyone else.

2.5 Item Response Theory

Classical test theory treats the total score as the unit of analysis and is silent about individual items. *Item response theory* (IRT) takes the complementary view, modelling the probability of each item response as a function of a respondent’s latent trait (Embretson & Reise, 2000; Hambleton et al., 1991). In the simplest model for a dichotomous item, the probability that a respondent with latent trait θ endorses item j is a logistic function of the gap between the trait and the item’s *difficulty* b_j , scaled by the item’s *discrimination* a_j ,

$$P(X_j = 1 \mid \theta) = \frac{1}{1 + e^{-a_j(\theta - b_j)}}. \quad (2.3)$$

The resulting S-shaped *item characteristic curve* rises from near zero for low-trait respondents to near one for high-trait respondents. IRT originates with the Rasch (1980) model (a one-parameter special case) and was developed for practical testing by Lord (1980); Edwards (2009) gives an accessible introduction for psychologists and Reise and Waller (2009) reviews its use in clinical measurement.

IRT matters here for one reason in particular: it furnishes a principled notion of an *aberrant* response pattern. Because the model specifies the probability of every response given the trait, it can quantify how *surprising* a respondent’s particular pattern is, and large surprise is a signature of misfit. This is the basis of the *person-fit* statistics (Meijer & Sijtsma, 2001) introduced in Chapter 1 as one family of careless-detection methods: a careless respondent answers in ways the measurement model deems improbable, and person-fit indices flag exactly that. ReReReRe pursues the same goal—finding respondents who violate the structure the rest of the sample obeys—but does so through cross-item correlation rather than a fitted IRT model, trading the model’s precision for robustness when the model is unknown or hard to estimate.

2.6 Latent Variables and Factor Analysis

Why should answers to different items correlate at all? The dominant answer in psychometrics is the *common-factor model*, which traces to Spearman (1904b) and was systematized by Thurstone (1947). The model posits that the observed responses are generated by a small number of unobserved *latent factors*—traits such as extraversion or conscientiousness—plus item-specific noise. Items that load on the same factor correlate because they share that common cause; items on different factors correlate weakly or not at all. Formally, the response of a respondent to item j is modelled as a weighted sum of the latent factors,

$$x_j = \lambda_{j1}f_1 + \lambda_{j2}f_2 + \cdots + \lambda_{jm}f_m + \varepsilon_j, \quad (2.4)$$

where $f_1, \dots, f_m$ are the latent factors, the *loadings* λ_{jk} quantify how strongly item j depends on factor k , and ε_j is the part of the item unique to itself (specific variance plus measurement error). The squared loadings summed across factors give the item’s *communality*, the proportion of its variance explained by the common factors.

Exploratory factor analysis (EFA) works backwards from the observed correlation matrix to estimates of the loadings and the number of factors, without assuming in advance which items load on which factor (Costello & Osborne, 2005; Fabrigar et al., 1999). It is “exploratory” in contrast to confirmatory factor analysis, where the loading pattern is specified ahead of time and tested. EFA matters for this thesis in two ways. The *generative* use comes first: the simulation studies of later chapters create artificial questionnaires from a known factor model of the form in Equation 2.4, so that the ground-truth structure is available for evaluation. The *analytic* use comes second: ReReReRe relies on the fact that attentive respondents reproduce the sample’s factor structure in their own answers, and one variant of the method uses EFA to choose which item pairs to score.

A practical question in any factor analysis is *how many factors to retain*. Retaining too few collapses distinct constructs; too many fits noise. Two classical heuristics are Cattell (1966)’s *scree test*, which looks for an “elbow” in the plot of successive eigenvalues, and the eigenvalue-greater-than-one rule. Both are now regarded as unreliable, and the recommended procedure is *parallel analysis* (Horn, 1965): one generates many random data sets with the same number of variables and respondents but no real factor structure, computes their eigenvalues, and retains only the factors whose eigenvalues in the real data exceed what random data produce by chance. Parallel analysis is, in effect, a permutation-style calibration of the factor count—a recurring logic in this thesis—and is the factor-retention method ReReReRe uses internally when it needs an estimate of dimensionality.

2.7 Reliability and Internal Consistency

Classical test theory defines reliability but does not say how to estimate it from a single administration. The most widely reported estimate is *coefficient alpha* (Cronbach, 1951), which uses the average inter-item correlation, adjusted for the number of items, to approximate the proportion of true variance in a multi-item scale; it ranges between 0 and 1, and high values indicate that the items “hang together” as they should if they measure a common construct.

Alpha is convenient but rests on a strong assumption—that all items are *essentially tau-equivalent*, contributing equal true-score information—that real scales rarely meet, in which case alpha can under- or over-estimate reliability. A substantial methodological literature now recommends *coefficient omega* (McDonald, 1999), which derives reliability from a fitted factor model and so does not require tau-equivalence (Revelle & Zinbarg, 2009; Sijsma, 2009). Recent tutorials advise reporting omega rather than alpha by default (Flora, 2020; Hayes & Coutts, 2020; McNeish, 2018); this thesis mentions alpha because it remains ubiquitous, but the argument it makes about careless responding is indifferent to which reliability coefficient is used.

A second classical result ties reliability directly to the correlations this thesis cares about. Spearman (1904a) showed that the correlation between two measured variables is *attenuated*—pulled toward zero—in proportion to their unreliability, and derived the *correction for attenuation* that recovers the disattenuated value. The consequence for careless responding is immediate: because careless answers add error variance (Section 2.4) and so lower reliability, they attenuate every correlation computed on the contaminated data. A study that fails to screen careless respondents does not merely add noise; it systematically understates the very relationships it set out to find.

Reliability thus connects directly to the detection problem. Because careless answers behave like added measurement error—raising the ε_j term of the factor model in Equation 2.4, lowering communalities, and weakening inter-item correlations—a drop in reliability when a scale is administered to a contaminated sample is one of the symptoms that motivate detection. Conversely, identifying and removing careless respondents typically *raises* reliability and sharpens the recovered factor structure, the practical payoff that Chapter 5 evaluates.

2.8 Validity

Reliability concerns whether a measurement is reproducible; *validity* concerns whether it measures what it claims to. The two are distinct—a scale can be highly reliable yet measure the wrong thing—and validity is the more fundamental requirement. The modern concept descends from Cronbach

and Meehl (1955), who introduced *construct validity*: the degree to which a score can be interpreted as reflecting the intended theoretical construct, established by accumulating evidence that the score behaves as the construct should. Messick (1995) recast validity as a single, unified, evidence-based judgement about the *interpretation and use* of scores rather than a property of the test in isolation, and Borsboom et al. (2004) pressed the stronger view that a test is valid only if variation in the attribute actually *causes* variation in the scores.

Careless responding is, on each of these framings, a direct threat to validity. When part of a sample answers without engaging item content, the scores no longer reflect the construct for those respondents, the evidential basis for any construct interpretation is undermined, and the causal link from attribute to response is broken. This is why data screening is not mere housekeeping but a precondition for valid inference (DeSimone et al., 2015): detecting and removing careless respondents protects the validity of every conclusion drawn from the remaining data. The detector developed in this thesis is, in this sense, an instrument for safeguarding construct validity.

2.9 Permutation and Randomization Tests

Classical hypothesis tests compare an observed statistic to a theoretical null distribution—the t , F , or χ^2 distribution—derived under assumptions such as normality. When those assumptions are doubtful, a *permutation* (or *randomization*) *test* offers an alternative that builds the null distribution *from the data themselves* (Good, 2005). The logic is simple and assumption-light: if the null hypothesis is true, then the observed arrangement of the data is just one of many equally likely arrangements obtainable by shuffling. One therefore repeatedly shuffles the data in a way consistent with the null, recomputes the statistic on each shuffle, and reads off where the observed value falls in the resulting distribution. A value far in the tail is unlikely under the null and constitutes evidence against it.

This is precisely the inferential principle behind ReReReRe. The method asks, of each respondent, whether their coherence across the item pairs that the sample deems related is higher than would be expected if their answers carried no such structure. The “no structure” null is realized by computing the same coherence over *randomly chosen* item pairs, repeated many times, to form a per-respondent null distribution. The respondent’s observed coherence is then expressed as a z -score relative to this distribution,

$$z = \frac{C_{\text{obs}} - \text{mean}(C_{\text{rand}})}{\text{sd}(C_{\text{rand}})}, \quad (2.5)$$

where C_{obs} is coherence over the genuinely coupled pairs and C_{rand} is the coherence over random pairs. A high z means the respondent is far more

coherent than chance—attentive; a low or negative z means they are no more coherent than random pairs would predict—potentially careless. Building the baseline per respondent makes the method *self-calibrating*: it does not assume any particular distribution for the coherence statistic, sidestepping exactly the normality problems that, as Chapter 1 noted, make the chi-square cutoff for Mahalanobis distance unreliable on Likert data.

2.10 The Bootstrap

A close cousin of the permutation test is the *bootstrap* (Efron, 1979; Efron & Tibshirani, 1993), a resampling method for quantifying the uncertainty of an estimate. Where a permutation test shuffles to simulate a null hypothesis, the bootstrap *resamples with replacement* to simulate the sampling variability of a statistic. Given a sample of n units, one draws many new samples of size n by sampling the original units with replacement (so some appear multiple times and others not at all), recomputes the statistic of interest on each resample, and uses the spread of these recomputed values as an estimate of the statistic’s sampling distribution.

The bootstrap is valuable precisely when a statistic is too complicated for a closed-form standard error. This thesis uses it for exactly such a case. To quantify how much ReReReRe contributes to the ensemble, Chapter 4 reports the *difference* in Matthews correlation coefficient between the full ensemble and an ablated ensemble without ReReReRe. No textbook formula gives the standard error of a difference of two MCCs, so a *percentile bootstrap* confidence interval is used: respondents are resampled with replacement several hundred times, the difference is recomputed on each resample, and the 2.5th and 97.5th percentiles of the resulting distribution form a 95% interval. If that interval excludes zero, the contribution is statistically distinguishable from no contribution—the form in which the thesis’s central ablation claim is stated.

2.11 Supervised Classification and the Random Forest

The ensemble at the heart of Chapters 4 and 5 is a *supervised classifier*: a model trained on examples whose true labels (careless or attentive) are known, which then predicts labels for new respondents. The specific classifier is the *Random Forest* (Breiman, 2001), built from *decision trees*. Because many psychologists know logistic regression but not tree ensembles, this section develops the Random Forest from the ground up; a fuller treatment appears in standard references (Hastie et al., 2009; James et al., 2013).

2.11.1 Decision trees

A *decision tree* is a sequence of yes/no questions about the predictor variables, arranged as a branching flowchart (Breiman et al., 1984). To classify a respondent, one starts at the top and answers each question—“is the ReReReRe z-score below -0.5 ?”, then “is the longstring above 12?”—following the corresponding branch until reaching a terminal *leaf*, which carries a predicted class (or class probability). The tree is *grown* from training data by repeatedly choosing, at each node, the question that best separates careless from attentive respondents according to a purity criterion such as the Gini index. A single tree has an appealing property—its decisions are transparent and can interact predictors automatically (the second question asked depends on the answer to the first)—but also a serious flaw: it is *high-variance*. Small changes in the training data can change which question is chosen near the top and thereby reshape the entire tree.

2.11.2 From one tree to a forest

The Random Forest tames this instability by averaging *many* de-correlated trees (Breiman, 2001). Two sources of randomness make the trees differ. First, each tree is grown on a *bootstrap resample* of the training respondents (the resampling idea of Section 2.10), so each tree sees a slightly different data set—a strategy called *bagging*, for bootstrap aggregating. Second, at each node only a *random subset* of the predictors is considered as candidate questions (typically about $\sqrt{p}$ of the p predictors), which prevents every tree from being dominated by the same one or two strong predictors. The forest’s prediction is the average of the predictions of its trees—in our setting, the proportion of trees that vote “careless,” interpreted as a predicted careless probability. Averaging many noisy but roughly unbiased trees reduces variance without inflating bias, which is why the forest predicts far more stably than any single tree.

2.11.3 Why a forest can beat logistic regression

Logistic regression models the log-odds of the outcome as a *linear, additive* function of the predictors. That is an excellent model when the truth is close to linear and additive, but careless detection is not such a case. The predictors interact: a respondent with both a very low ReReReRe z-score *and* an unusually low intra-individual response variability is far more suspicious than the additive combination of the two signals would imply, because that conjunction is the signature of straightlining. A Random Forest captures such interactions automatically through the nested structure of its splits, without the analyst having to specify interaction terms in advance. In the simulation studies of later chapters, the forest outperforms logistic regression

Table 2.1: The confusion matrix for careless detection. “Positive” denotes a respondent flagged as careless.

	Truly careless	Truly attentive
Flagged careless	True positive (TP)	False positive (FP)
Flagged attentive	False negative (FN)	True negative (TN)

on every evaluation metric, with the largest margins on the hardest cases (short questionnaires and low careless base rates).

A further benefit is *variable importance*: by measuring how much predictive accuracy degrades when a predictor is randomly permuted, a Random Forest ranks the predictors by their contribution (Strobl et al., 2008). This provides a principled way to ask which detectors carry the signal—an analysis the thesis uses to show that ReReReRe and a small set of auxiliary indices account for most of the ensemble’s power.

2.12 Cross-Validation

A classifier evaluated on the same data used to train it will look better than it really is, because it can memorize idiosyncrasies of the training sample that do not generalize—a problem called *optimism* or *overfitting*. *Cross-validation* estimates out-of-sample performance without requiring a separate test set (Kohavi, 1995; Stone, 1974). In *k-fold* cross-validation, the data are partitioned into k roughly equal folds (here $k = 5$). The model is trained on $k - 1$ folds and used to predict the held-out fold; this is repeated k times so that each fold serves once as the held-out set. Every respondent thus receives a prediction from a model that did *not* see them during training. These *out-of-fold* predictions are concatenated and used for all performance estimates, giving an honest picture of how the classifier would behave on new respondents drawn from the same population. The folds are *stratified*—constructed to preserve the careless base rate within each fold—so that no fold is unrepresentative.

2.13 Evaluating a Binary Classifier

Once a classifier assigns each respondent a predicted careless probability, turning those probabilities into a binary decision requires a *threshold*, and judging the decisions requires *metrics*. Both rest on the *confusion matrix* (Table 2.1), which cross-tabulates the predicted label against the truth into four counts: true positives (TP, careless correctly flagged), true negatives (TN), false positives (FP, attentive respondents wrongly flagged), and false negatives (FN, careless respondents missed).

Table 2.2: Principal evaluation metrics derived from the confusion matrix.

Metric	Definition	Range
Sensitivity (recall)	$TP/(TP + FN)$	0 to 1
Specificity	$TN/(TN + FP)$	0 to 1
Precision (PPV)	$TP/(TP + FP)$	0 to 1
F1	harmonic mean of precision and recall	0 to 1
MCC	Equation 2.6	-1 to +1

From these four counts a family of metrics is derived. *Sensitivity* (recall, true-positive rate) is the proportion of truly careless respondents that are flagged, $TP/(TP + FN)$; *specificity* is the proportion of attentive respondents correctly left unflagged, $TN/(TN + FP)$. *Precision* (positive predictive value) is the proportion of flagged respondents who are genuinely careless, $TP/(TP + FP)$. The *F1 score* is the harmonic mean of precision and recall. These are summarized in Table 2.2.

The thesis adopts the *Matthews correlation coefficient* (MCC) as its primary metric (Chicco & Jurman, 2020; Matthews, 1975). The MCC is, in effect, the Pearson correlation between the predicted and the true binary labels,

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}, \quad (2.6)$$

ranging from -1 (perfect disagreement) through 0 (chance) to $+1$ (perfect agreement). Its decisive advantage over accuracy and F1 is that it uses all four cells of the confusion matrix and is therefore robust to *class imbalance*, which careless detection always involves: when only, say, 10% of respondents are careless, a useless classifier that flags no one achieves 90% accuracy, and even F1 can be inflated, whereas the MCC of such a classifier is 0 (Chicco & Jurman, 2020). A high MCC requires the classifier to do well on *both* classes simultaneously.

Two *threshold-free* metrics complete the picture by summarizing performance across all possible thresholds at once. The *area under the receiver operating characteristic curve* (AUC) is the probability that a randomly chosen careless respondent receives a higher predicted probability than a randomly chosen attentive one (Hanley & McNeil, 1982); 0.5 is chance and 1 is perfect. The *area under the precision-recall curve* (AUPRC) is a companion measure that, by ignoring the typically very large true-negative cell, is more informative than AUC under heavy class imbalance (Saito & Rehmsmeier, 2015). Because AUC and AUPRC do not depend on a chosen cutoff, they isolate the quality of the underlying ranking from the separate question of where to set the threshold.

That separate question—*which* threshold to use—is itself consequential, as Chapter 1 stressed. A threshold defines an *operating point*, a specific trade-off between sensitivity and specificity; moving it trades missed careless respondents against falsely flagged attentive ones. Chapter 5 returns to the problem of choosing an operating point without access to ground-truth labels, which is the form the threshold problem takes in real deployment.

2.14 Summary

The pieces assembled here recur throughout the thesis as a connected toolkit rather than a list. Correlation, computed at the level of the individual rather than the sample, is what ReReReRe measures; the common-factor model explains why that correlation exists for attentive respondents and parallel analysis estimates its dimensionality; the permutation test turns the measurement into a self-calibrating per-respondent z-score; the bootstrap puts a confidence interval on the detector’s contribution; the Random Forest combines that detector with auxiliary indices, cross-validation estimates the combination’s performance without optimism, and the Matthews correlation coefficient—together with AUC and AUPRC—reports how good that performance is under the class imbalance intrinsic to careless detection. With these concepts in place, Chapter 3 can introduce the ReReReRe detector itself.

Chapter 3

ReReReRe: A Permutation-Based Coherence Detector

3.1 Motivation and Core Idea

Chapter 1 closed on a gap. The detection methods in widest use are strongest on *consistent* careless responding—straightlining, acquiescence, machine-like regularity—and weakest on *inconsistent* careless responding, where a respondent’s answers vary across items but bear no coherent relationship to item content. Random and mixed responding are the hardest patterns to prevent and the most damaging to covariance-based analyses, yet the only established index that targets them directly, the psychometric-synonym statistic (Meade & Craig, 2012), operates on a handful of hand-selected item pairs. This chapter introduces *ReReReRe*, a detector built specifically to exploit the full correlational structure of a multi-construct questionnaire in order to identify inconsistent careless responding.

The core idea is a single observation made operational. An attentive respondent’s answer to one item is informative about their answer to a correlated item, because both items load on a shared latent factor (Section 2.6). A careless respondent’s answers to the same pair are *incoherent*: the cross-item structure that the sample as a whole exhibits is absent in their individual profile. *ReReReRe* measures, for each respondent, how much of the sample’s pairwise structure their own answers reproduce, and compares that quantity against a baseline obtained by permutation. The result is a per-respondent z-score: high when the respondent is more coherent than chance (attentive), low or negative when they are no more coherent than randomly paired items would predict (potentially careless). Because the baseline is built separately for each respondent, the method is *self-calibrating*: it makes no distributional assumption about the coherence statistic, sidestepping the normality prob-

lems that, as Chapter 1 noted, cripple the chi-square cutoff for Mahalanobis distance on Likert data.

3.2 The Algorithm

3.2.1 Individual-level coherence

Let X be the $N \times J$ matrix of responses (N respondents, J items). The sample-level correlation matrix R summarizes how items relate across respondents. ReReReRe selects a set of k item pairs $\{(a_1, b_1), \dots, (a_k, b_k)\}$ that the sample deems related (Section 3.2.2), and then, for each respondent r , forms the two vectors of that respondent's answers to the left-hand and right-hand members of the pairs,

$$\mathbf{u}_r = (X_{r,a_1}, X_{r,a_2}, \dots, X_{r,a_k}), \quad \mathbf{v}_r = (X_{r,b_1}, X_{r,b_2}, \dots, X_{r,b_k}).$$

The respondent's *coherence* is a weighted association between $\mathbf{u}_r$ and $\mathbf{v}_r$, with each pair contributing in proportion to its sample-level $|r|$,

$$C_{\text{obs}}(r) = \sum_{j=1}^k w_j \tilde{u}_{r,j} \tilde{v}_{r,j}, \quad w_j = \frac{|R_{a_j b_j}|}{\sum_l |R_{a_l b_l}|}, \quad (3.1)$$

where $\tilde{u}, \tilde{v}$ denote the item-standardized values. Strong pairs count more than weak ones, but no pair is discarded outright. The crucial point is that Equation 3.1 is computed *down a respondent's row*, not across respondents: it asks whether *this person* answered the coupled pairs coherently.

3.2.2 Pair selection: two regimes

How the k pairs are chosen depends on questionnaire length, because the signal ReReReRe relies on comes from the total number of high-correlation pairs available, and that number scales very differently for short and long instruments. ReReReRe therefore switches automatically between two regimes at a threshold of 60 items (the `mode = "auto"` default).

Weighted regime (used for $J \leq 60$ items). Every one of the $\binom{J}{2}$ item pairs is included, each weighted by its absolute sample correlation as in Equation 3.1. With short questionnaires there are too few strongly correlated pairs to select a clean subset, so discarding any of them throws away signal; using all pairs but down-weighting the weak ones extracts the most from limited structure.

Coupled regime (used for $J > 60$ items). Only the top `corProp` fraction of pairs by absolute correlation is retained (default 3%), subject to a floor of `min_pairs = 15`. With long questionnaires the strongest pairs carry a clean, redundant signal, and restricting attention to them sharpens the contrast

between attentive and careless respondents while keeping the computation tractable.

The switch is automatic and recorded in the output (`mode_used`); a user can override it to force either regime. Calibration studies confirm a crossover near 60 items: below it the weighted regime wins by up to 0.06 MCC, above it the coupled regime wins by a similar margin.

3.2.3 The permutation baseline and the z-score

A high coherence is only meaningful relative to what coherence the same respondent would show over pairs with *no* shared structure. ReReReRe constructs this baseline by permutation (Section 2.9). For each of `iterations` = 100 rounds, it draws k *random* item pairs from the matrix, recomputes the same weighted coherence as in Equation 3.1, and thereby obtains, for each respondent, a null distribution $\{C_{\text{rand}}(r, 1), \dots, C_{\text{rand}}(r, 100)\}$. The respondent’s score is the standardized distance of their observed coherence from this personal baseline,

$$z_{\text{RR}}(r) = \frac{C_{\text{obs}}(r) - \text{mean}_s C_{\text{rand}}(r, s)}{\text{sd}_s C_{\text{rand}}(r, s)}. \quad (3.2)$$

This is the permutation z-score of Equation 2.5, specialized to coherence. A respondent is flagged as careless when z_{RR} falls below a threshold (Section 3.5). The algorithm is summarized in the box below.

Algorithm: ReReReRe (standard mode).

1. Rescale each item (default: divide by its observed maximum; Section 3.4).
2. Compute the sample correlation matrix R ; record the sign of each correlation.
3. *Sign alignment*: for each pair with $R_{ab} < 0$, reverse-code column b (Section 3.3).
4. Select pairs: all pairs (weighted, $J \leq 60$) or the top- $k\%$ by $|r|$ (coupled, $J > 60$).
5. Compute each respondent’s weighted coherence C_{obs} (Eq. 3.1).
6. Repeat 100 times: draw k random pairs, sign-align, compute C_{rand} .
7. Convert to a per-respondent z-score z_{RR} (Eq. 3.2).
8. Flag respondents with $z_{\text{RR}} \leq \text{threshold}$ (default 1.5, or auto-calibrated).

In R the entire procedure is a single call, `ReReReRe(data)`, returning a data frame with each respondent’s z_{RR} , the flag, and metadata (the regime used, the calibrated threshold, the estimated number of factors). The default parameters are listed in Table 3.1.

Table 3.1: Default parameters of `ReReReRe()` and their role.

Parameter	Default	Role
<code>mode</code>	"auto"	Weighted if $J \leq 60$, coupled if $J > 60$
<code>corProp</code>	0.03	Fraction of top- $ r $ pairs kept (coupled regime)
<code>min_pairs</code>	15	Floor on the number of coupled pairs
<code>iterations</code>	100	Number of permutation rounds for the baseline
<code>align_signs</code>	TRUE	Reverse-code negatively-correlated pairs
<code>rescale</code>	"proportion"	Per-item rescaling for mixed scales
<code>z_threshold</code>	1.5	Flagging cutoff (or <code>auto</code>)
<code>auto_z</code>	FALSE	Calibrate the cutoff from total item count

3.3 Reverse-Coded Items: Sign Alignment

Questionnaires routinely include *reverse-keyed* items, worded so that agreement indicates *less* of the trait. Such an item correlates *negatively* with same-trait items keyed in the usual direction. This creates a subtle but fatal problem for any method that aggregates correlations across pairs. If the coherence of Equation 3.1 mixes positively-keyed pairs (which contribute positive products for an attentive respondent) with negatively-keyed pairs (which contribute negative products), the two cancel inside the sum, and even a perfectly coherent respondent receives $C_{\text{obs}} \approx 0$. Every respondent then looks equally incoherent, and the z-score carries no signal.

The defect is not hypothetical. It was discovered on the Robie HEXACO-60 data, where 52% of the coupled pairs were oppositely keyed: with the pairs left unaligned, every respondent received $z_{\text{RR}} \approx 0$ (mean observed coherence 0.078 versus a baseline mean of 0.084), and the detector’s area under the curve sat at chance. The fix, controlled by `align_signs` (default TRUE), is to use the *sign* of each pair’s sample-level correlation to align the pair before computing coherence: wherever $R_{ab} < 0$, column b is reverse-coded (mapped by $x \mapsto (\max_b + \min_b) - x$) so that the pair contributes positively for a coherent respondent. The same alignment is applied to the random pairs of the permutation baseline, so the null distribution is comparable. With the fix in place on the Robie data, mean coherence rose to 0.299, the mean z-score to 3.49, and the area under the curve from 0.47 to 0.66.

Sign alignment is more than data hygiene: it is what lets ReReReRe operate on real questionnaires *without* requiring the user to recode reverse-keyed items by hand. The procedure is transparent and emits a warning only when it is disabled in the presence of negatively-keyed pairs.

3.4 Mixed Response Scales: Rescaling

A second real-data complication is *heterogeneous response scales*: a battery that mixes, say, 1–4 and 1–7 items. Because the Pearson correlation in Equation 3.1 is sensitive to the numerical spacing of the response options, longer-range items would otherwise dominate a respondent’s coherence simply by virtue of their larger spread. ReReReRe addresses this with an optional per-item rescaling step (**rescale**), applied before any correlation is computed.

The default, **"proportion"**, divides each item by its observed maximum. This choice is principled. Section 2.2 noted that the Pearson correlation is invariant to identical affine transforms applied to all items: on a *homogeneous*-scale questionnaire every column is divided by the same constant, so the coherence is provably unchanged—the rescaling costs nothing. On a *heterogeneous*-scale questionnaire it equalizes the contribution of items measured on different ranges. On a stress test of 80 items split between 1–4 and 1–7 scales, proportion rescaling lifted MCC from 0.32 (no rescaling) to 0.43, while leaving uniform-scale results numerically identical. More aggressive options are exposed for advanced users—**"minmax"** and **"zscore"**—and **"none"** reproduces the legacy behaviour while warning when item ranges are heterogeneous. The recommendation is nonetheless to keep response scales uniform where possible; rescaling is a safeguard, not a substitute for clean design.

3.5 Threshold Selection and Auto-Calibration

The z-score of Equation 3.2 is continuous; flagging requires a cutoff. Two findings make this less fraught than it sounds. First, the method is *not* sharply threshold-sensitive: across a wide range of conditions, any cutoff between roughly 1.0 and 2.0 performs near-optimally, with a flat plateau of detection quality. The recommended universal default is $z_{\text{threshold}} = 1.5$, interpreted as “the respondent’s coherence is 1.5 standard deviations above their own random baseline.”

Second, where one wants to do better than a universal default, the optimal cutoff has a clean and interpretable dependence on questionnaire size. Figure 3.1 shows the optimal threshold as a function of the number of factors, holding items per factor fixed. The curve is *U-shaped*: at very low factor counts the z-scores of attentive and careless respondents overlap heavily, forcing a low threshold to catch anyone; as the number of factors grows the two distributions separate, and a *higher* threshold becomes optimal because it buys specificity without sacrificing sensitivity. The blue point cloud also encodes detection quality through marker size, which grows visibly from left (small, $\text{MCC} < 0.15$) to right (large, $\text{MCC} > 0.5$): more factors mean both a higher optimal threshold and better detection.

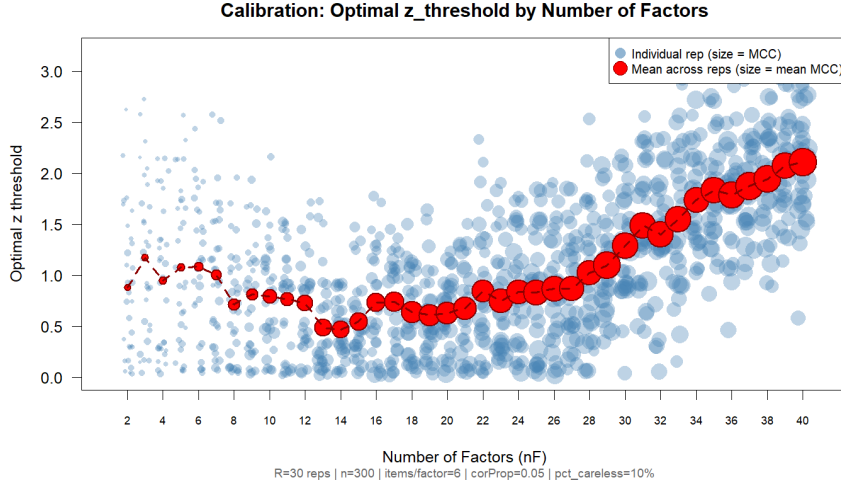


Figure 3.1: Optimal z -threshold as a function of the number of factors (items per factor fixed at 6, $n = 300$, 10% careless, 30 replications). Blue points are individual replications (size $\propto$ MCC); red points are replication means. The optimal threshold follows a U-shape, reaching a minimum near $z \approx 0.5$ at 13–15 factors and rising to $z \approx 2.1$ at 40 factors.

Rather than calibrate on the number of factors alone, the implementation uses the better predictor identified by a two-dimensional calibration study: the *total* number of items. A simple linear rule, $z \approx 0.505 + 0.0042 \times (\text{total items})$, explains roughly twice the variance of a rule based on the factor count, because what governs the separation of the z -score distributions is the total number of high-correlation pairs available, which scales with both the number of factors and the items per factor. Setting `auto_z = TRUE` activates this calibration: ReReReRe estimates the number of factors by parallel analysis (Section 2.6), looks up the calibrated threshold, and applies it. In aggregate, auto-calibration performs about as well as the fixed $z = 1.5$ default, confirming that the method does not demand careful tuning; its value is convenience, removing a parameter choice for non-technical users.

3.6 Intrinsic Properties

Before comparing ReReReRe to other detectors, it is worth characterizing how its performance depends on questionnaire structure, because those dependencies explain both its strengths and its limits. The picture below comes from a large simulation multiverse crossing the number of factors, items per factor, sample size, and careless base rate, with the structure of each dataset known by construction.

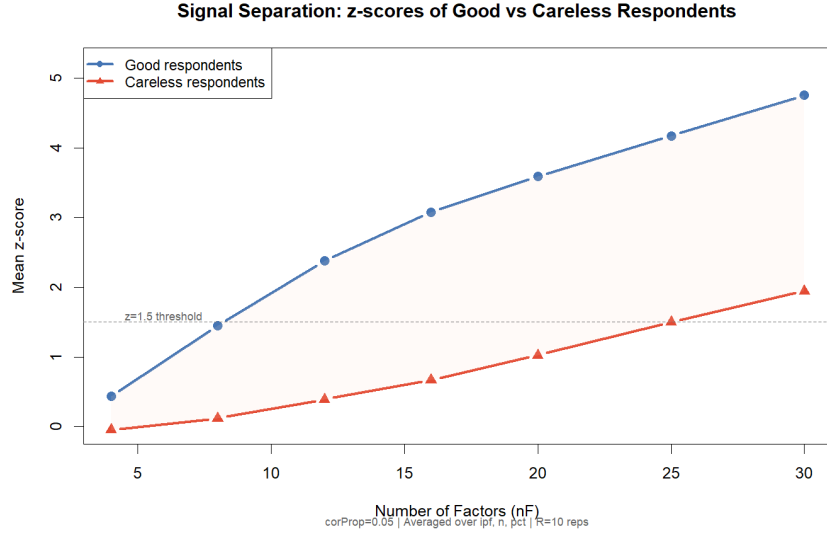


Figure 3.2: Mean z-score of attentive (“good”) versus careless respondents as a function of the number of factors (averaged over items per factor, sample size, and base rate; 10 replications). The separation grows quasi-linearly with dimensionality, explaining the method’s scaling advantage. The dashed line marks the $z = 1.5$ default cutoff.

3.6.1 Signal grows with dimensionality

The mechanism behind ReReReRe predicts that more factors should yield a cleaner separation between attentive and careless respondents, because a richer factor structure offers more coherent pairs against which incoherence stands out. Figure 3.2 confirms this directly. It plots the mean z-score of attentive and of careless respondents as the number of factors increases. At four factors the two groups overlap almost completely (a gap of about 0.48); by thirty factors they are separated by nearly three standard deviations (a gap of about 2.81). The attentive curve climbs steeply while the careless curve stays low, and the $z = 1.5$ default sits cleanly between them across most of the range.

3.6.2 Total length is the unifying predictor

Decomposing the variance in detection quality, the two strongest drivers are the items per factor ($\eta^2 = 0.31$) and the number of factors ($\eta^2 = 0.26$); sample size and base rate matter much less. Crucially, neither factor count nor items per factor alone is the right summary—their *product*, the total questionnaire length, is. Figure 3.3 plots detection quality against total items and collapses the separate effects onto a single saturating curve: MCC rises steeply up to roughly 150 items and then plateaus, with a slight decline

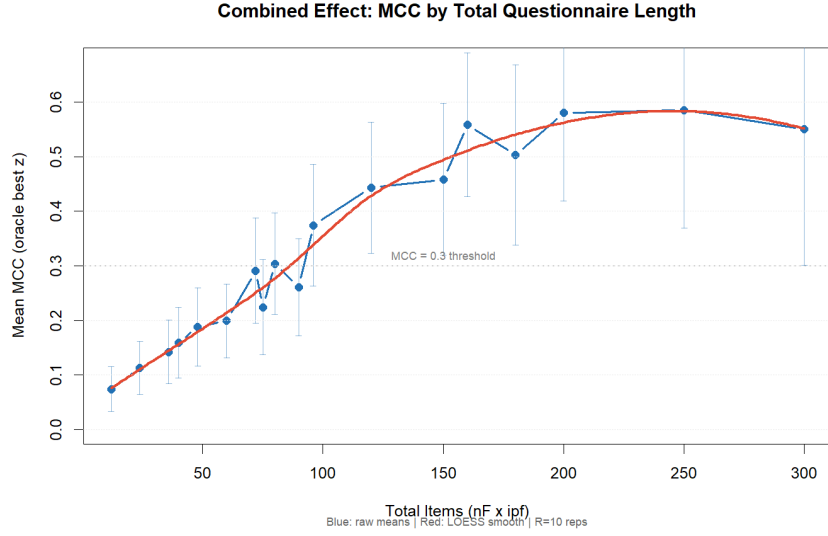


Figure 3.3: Detection quality (oracle-best MCC) as a function of total questionnaire length (number of factors $\times$ items per factor). Blue points are condition means with ± 1 SD bars; the red curve is a LOESS smooth. MCC saturates above roughly 150 items.

beyond about 250 items where the number of items begins to approach the number of respondents and individual-level correlations grow noisy. The practical reading is that ReReReRe should be described in terms of *total length*, not number of constructs: a 120-item questionnaire of 12 factors with 10 items each detects far better than a 75-item questionnaire of 25 factors with 3 items each, despite the latter’s higher factor count.

3.6.3 Detection by careless pattern

Broken down by the type of injected careless responding, ReReReRe detects all three inconsistent patterns at comparable rates: longstring at about 83%, mixed at about 81%, and random at about 79% (averaged over the favorable threshold range). This even profile is expected, since all three corrupt the cross-pair structure the method monitors. The limits of the method, examined next, lie not in any one inconsistent pattern but in the *consistent* patterns it is not designed to see.

3.7 Benchmark Comparison: ReReReRe versus Mahalanobis

Mahalanobis distance is the most widely used multivariate-outlier index for careless detection (Section 1.4.2), and it is the natural benchmark for ReReReRe. The comparison below was run on *identical* simulated datasets, so the two detectors face exactly the same data. Two framings are necessary, because the comparison hinges on how each method’s threshold is set.

3.7.1 Practical thresholds: the decisive contrast

In practice, Mahalanobis distance is applied with a fixed cutoff: respondents whose squared distance exceeds the chi-square critical value at $\alpha = .001$, with degrees of freedom equal to the number of items, are flagged (Curran, 2016; Meade & Craig, 2012). This is the Mahalanobis counterpart of ReReReRe’s $z = 1.5$ default—a rule requiring no access to labels. Under this realistic, label-free comparison the two methods are not close. The chi-square rule is *essentially non-functional* on Likert data: its sensitivity is between 0.4% and 1.9%, so it catches almost no careless respondents, for an MCC of about 0.06. ReReReRe with its fixed default reaches an MCC of about 0.27 over the same conditions—more than four times higher. The reason is exactly the one developed in Chapter 1: the chi-square reference distribution assumes multivariate normality, which Likert data violate badly, so the nominal cutoff is far too conservative. ReReReRe’s permutation baseline makes no such assumption.

3.7.2 Oracle thresholds and the threshold-free view

It might be objected that Mahalanobis distance could do better with a tuned cutoff. This is true, but the tuning requires the true labels—an *oracle* no analyst possesses. Even granting Mahalanobis its oracle-best threshold per condition, ReReReRe (also at its oracle best) overtakes it as dimensionality grows: the two are tied near ten factors ($\text{MCC} \approx 0.37$ for both), after which ReReReRe pulls ahead, reaching $\text{MCC} \approx 0.57$ at twenty-five factors versus ≈ 0.39 for Mahalanobis, whose performance stays roughly flat. The fairest single comparison is the threshold-free area under the curve, which removes the thresholding question entirely (Section 2.13). Here ReReReRe’s AUC, computed with fully fixed defaults, climbs from 0.59 at four factors to 0.84 at twenty-five, overtaking Mahalanobis (flat at about 0.64) from eight factors upward. The conclusion is consistent across framings: Mahalanobis is a slightly better detector for very short questionnaires but does not scale, whereas ReReReRe scales strongly with questionnaire length and, under any realistic thresholding, dominates by a wide margin.

The two methods detect different things, and this is the deeper point.

Mahalanobis distance flags multivariate *outliers*—respondents far from the sample centroid—whether careless or merely unusual. ReReReRe flags *incoherence*—a breakdown of the cross-factor structure. The next chapter exploits this complementarity rather than choosing between the two.

3.8 Design Alternatives Explored

The canonical `ReReReRe()` described above—the weighted/coupled auto-switch with sign alignment and proportion rescaling—was selected after a broad search over alternative formulations. They are noted here briefly for completeness; none displaced the canonical method as the recommended detector.

Factor-guided pair selection (an EFA-based variant, exposed as `ReReReRe_F()`) chooses within-factor pairs from an exploratory factor solution rather than by raw correlation. It wins on *simulated* data, where the factor structure is clean, but loses on *real* validation datasets, where parallel analysis on noisy, cross-loaded data produces poor factor assignments; the empirical top- $|r|$ ranking of the coupled regime proved more robust to contamination. *Iterative* re-estimation (re-running the factor solution on the respondents judged attentive, then re-scoring) gave a small improvement on mid-length questionnaires but not elsewhere. A *split-half* scorer and a *variance penalty* (which pushes near-zero-variance straightliners toward negative z) each helped on specific partial-corruption or consistent-careless cases but did not improve overall classification enough to justify added complexity. The lesson from these experiments—that a method’s edge on simulated data need not survive on real data—shaped the validation philosophy of the chapters that follow. The variant code remains in the software for reference but is not part of the recommended pipeline.

3.9 Limitations and the Case for an Ensemble

ReReReRe is a strong standalone detector of inconsistent careless responding, but it has three limits, and naming them sets up the next chapter. First, it is *weak at low dimensionality*: with few factors the z -score distributions of attentive and careless respondents overlap (Figure 3.2), and detection is modest below roughly eight factors. Second, by construction it targets *inconsistent* careless responding and adds little on *consistent* patterns—pure straightlining and acquiescence—which a simple longstring or low-variability rule already catches perfectly. Third, even at its best a single threshold leaves a gap to the oracle, because no label-free rule can perfectly separate the two classes.

Each limit is the mirror image of a strength of some other detector. Mahalanobis distance and longstring catch the consistent patterns ReReReRe

misses; low-variability indices catch straightlining outright. This complementarity suggests that the right way to use ReReReRe is not alone but as one informative feature among several, combined by a classifier that learns when to trust each. That is the subject of Chapter 4, which embeds ReReReRe in a Random Forest ensemble alongside the auxiliary detectors and quantifies, with the confidence intervals introduced in Section 2.10, exactly how much the coherence signal contributes.

Chapter 4

An Ensemble Detector

4.1 From One Detector to Many

Chapter 3 ended on a diagnosis: ReReReRe is a strong detector of *inconsistent* careless responding but is, by design, nearly blind to *consistent* patterns such as straightlining and acquiescence, weak at low dimensionality, and—like any single index—limited by the need to commit to one threshold. Each of these limits is the mirror image of a strength of some other detector. Longstring and low intra-individual variability catch straightliners outright; Mahalanobis distance catches multivariate outliers. The detectors are *complementary*: they fail on different respondents.

This chapter turns that complementarity into a single, better detector. Rather than choosing one index or applying several under separate ad hoc cutoffs—the informal multi-index practice that Chapter 1 described as the field’s de facto recommendation (Curran, 2016; Ward & Meade, 2023)—it treats each index as one *feature* and lets a trained classifier learn how to weigh them and when to trust each. The classifier is a Random Forest (Breiman, 2001), and the central empirical question of the chapter is whether ReReReRe contributes anything *beyond* the established auxiliary detectors, and if so, how much and under what conditions. The answer, developed with bootstrap confidence intervals, is that its contribution is statistically positive at every questionnaire length and grows substantially with length.

4.2 The Five Features

Each respondent is summarized by five detector scores, all oriented so that *higher means more careless*:

- z_{RR} the ReReReRe permutation coherence z-score of Chapter 3, sign-flipped so that low coherence (incoherent, inconsistent careless) reads as a high feature value.

- IRV** the (negated) intra-individual response variability (Dunn et al., 2018): low dispersion—the signature of straightlining—reads as high careless.
- LongString** the longest run of identical consecutive responses (Johnson, 2005).
- D^2 the Mahalanobis distance from the sample centroid (Mahalanobis, 1936): multivariate outlyingness.
- Person–Total** the (negated) correlation between a respondent’s profile and the sample mean profile: answering against the grain.

The first feature is the contribution of this thesis; the remaining four are the established indices reviewed in Chapter 1. The design intention is explicit division of labour: ReReReRe carries the *inconsistent*-careless signal that the four auxiliaries miss, while the auxiliaries carry the *consistent* signal that ReReReRe misses. Whether that intention is realized is tested in Section 4.6.

4.3 The Random Forest Combiner

The five features are combined by a Random Forest (Section 2.11): an ensemble of 200 decision trees, each grown on a bootstrap resample of the training respondents, each split considering a random subset of the features, the final predicted careless probability being the proportion of trees voting “careless.” Two properties make the Random Forest the right combiner here. First, the features interact: a respondent with both a low z_{RR} and a low IRV is far more suspicious than the additive combination of the two signals implies, because that conjunction is the signature of a straightliner who also breaks cross-factor coherence. Trees capture such interactions automatically. Second, the Random Forest is robust to features on different scales and to redundancy among them, so the five detectors can be fed in raw. In our simulations the Random Forest outperformed logistic regression and elastic-net regression on every evaluation metric considered, with the widest margins on the hardest conditions (short questionnaires and low base rates), consistent with the non-linear, interactive structure of the problem.

Throughout, the classifier is evaluated by five-fold cross-validation (Section 2.12): each respondent receives an out-of-fold predicted probability from a forest that did not see them in training, and these out-of-fold predictions are used for every metric reported below. This guards against the optimism that would arise from scoring the classifier on its own training data.

4.4 Operational Definition of Careless

Evaluating a detector requires a ground-truth label, and in simulated data a respondent’s corruption can be set to any degree. We therefore need an

explicit rule for *how corrupted* a respondent must be to count as careless. This rule is not innocuous: set it too high and the negative class is contaminated with heavily corrupted respondents; set it too low and trivially-corrupted respondents inflate the positive class.

We fix the rule empirically. A respondent is labelled careless when the corrupted fraction of their row exceeds $\tau = 0.40$, and *every* other respondent—clean and lightly corrupted alike—stays in the negative class. The cutoff $\tau = 0.40$ is the joint argmax of the Matthews correlation coefficient, Cohen’s κ , Youden’s J, and balanced accuracy across a sweep $\tau \in \{0.10, 0.20, \dots, 0.90\}$ evaluated on 24,000 simulated respondents. (Two secondary metrics, F1 and AUPRC, favour slightly lower cutoffs of 0.30 and 0.20; the four primary classification metrics agree unanimously on 0.40.) Crucially, this is the *deployment-realistic* framing: in real data the analyst does not know in advance who has 30% versus 60% corruption, so keeping the lightly-corrupted respondents in the negative class tests whether the detector recovers moderately-to-fully corrupted rows without false-flagging the ambiguous middle. An earlier, looser definition that excluded the middle from the negative class was discarded as artificially easy.

4.5 Simulation Design and Headline Performance

The ensemble was evaluated on a simulation grid of **384 datasets** crossing eight questionnaire sizes (30, 50, 80, 100, 150, 200, 250, and 300 items), four careless base rates (10%, 20%, 40%, 60%), and twelve replications per cell, with $n = 500$ respondents each. Within every dataset, careless respondents were injected at one of six prototypical patterns—random, longstring, mixed, fatigue, acquiescent, pure straightlining—with the corrupted fraction drawn from $\{0\%, 10\%, \dots, 100\%\}$, so that the full corruption continuum is represented. The five features were computed for every respondent and a Random Forest trained under five-fold cross-validation.

The operating point is set by the *rate-aware* calibration, the deployable rule developed in Chapter 5: the analyst supplies a rough prior on the careless base rate and the threshold is placed at the corresponding quantile of the predicted probabilities, requiring no labels. Pooled across all sizes and rates, the ensemble reaches the performance in Table 4.1.

Detection quality scales smoothly with questionnaire length (Table 4.2). At 30 items the ensemble is useful but modest ($\text{MCC} = 0.44$); by 150 items it is strong ($\text{MCC} = 0.72$); and it approaches its asymptote near 300 items ($\text{MCC} = 0.82$), where the $250 \rightarrow 300$ gain is only $+0.017$. This is the same total-length scaling identified for ReReReRe alone in Chapter 3, now inherited by the ensemble.

Table 4.1: Pooled detection performance of the five-feature ensemble under the rate-aware operating point (160 cells, five-fold CV within each).

Metric	Pooled value
MCC	0.662
F1	0.730
Sensitivity	0.731
Specificity	0.932
AUC	0.919

Table 4.2: Per-size detection (MCC) under the rate-aware operating point.

Items	30	50	80	100	150	200	250	300
MCC	0.439	0.525	0.604	0.630	0.717	0.771	0.798	0.815

4.6 The Contribution of ReReReRe

The chapter’s central question is whether ReReReRe earns its place in the ensemble. We answer it by *ablation*: training the same Random Forest under two feature sets—the full five features, and the four auxiliaries only—and comparing them under the identical cross-validation protocol and operational definition.

4.6.1 Ablation by questionnaire length

Figure 4.1 shows the two configurations as a function of questionnaire length. The full ensemble (with ReReReRe) rises from $\text{MCC} \approx 0.42$ at 30 items to ≈ 0.74 at 300; the auxiliaries-only ensemble rises from ≈ 0.40 to only ≈ 0.61 and then plateaus. The gap between the curves *widens monotonically* with length: at short questionnaires the four auxiliaries already capture most of the available signal, but at long questionnaires they saturate while the full ensemble keeps improving. Adding more auxiliary detectors does not close this gap—it is specifically the cross-factor coherence signal, available only from ReReReRe, that the auxiliaries cannot supply.

4.6.2 Bootstrap confidence intervals on the contribution

A difference of curves is suggestive; to make the claim inferential we put a confidence interval on it. For each questionnaire size we bootstrapped over respondent indices (500 resamples; Section 2.10) and recomputed $\Delta\text{MCC} = \text{MCC}(\text{full}) - \text{MCC}(\text{auxiliaries only})$ on each resample, taking the 2.5th and 97.5th percentiles as a 95% interval. Table 4.3 and Figure 4.2 report the result.

The interval *excludes zero at every size tested*, so the contribution of

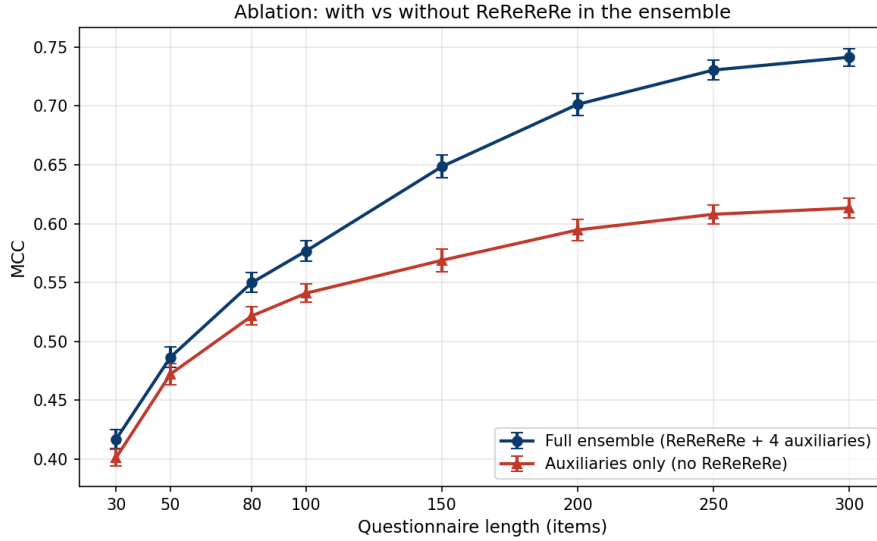


Figure 4.1: Ablation by questionnaire length. The full five-feature ensemble (ReReReRe + four auxiliaries) versus the four auxiliaries alone, both trained under five-fold cross-validation with the same operational definition. Error bars are ± 1 SEM. The gap grows from a small margin at 30 items to a sizeable one at 200–300 items.

ReReReRe is statistically positive throughout—including the 30-item case, where one might have expected it to be redundant. The *magnitude*, however, is strongly length-dependent: small below 100 items ($\approx +0.02$ to $+0.035$) and large above 150 items ($+0.10$ to $+0.15$, roughly a quarter of the auxiliaries’ ceiling). This is the chapter’s headline structural finding, and it has a clear practical reading: ReReReRe pays off most on the long multi-construct batteries that dominate modern psychometric practice (150–300 items), not on the short unidimensional scales where it would be easiest to dismiss as redundant.

4.7 Where the Contribution Comes From

The ablation says *how much* ReReReRe contributes; the per-pattern breakdown says *where*. Figure 4.3 plots detection rate against corruption fraction for each of the six injected patterns, comparing the full ensemble with the auxiliaries-only ensemble.

The picture is exactly the complementarity the ensemble was designed to exploit. On the *consistent* patterns—pure straightlining, acquiescence, and highly-corrupted longstring and fatigue—the two curves are essentially identical: the auxiliary detectors already saturate, and ReReReRe adds

Table 4.3: Bootstrap 95% confidence intervals on the contribution of ReReReRe, $\Delta\text{MCC} = \text{MCC}(\text{full}) - \text{MCC}(\text{auxiliaries only})$, by questionnaire length.

Items	ΔMCC	95% CI	$P(\Delta > 0)$
30	+0.020	[+0.009, +0.031]	1.000
50	+0.018	[+0.008, +0.028]	1.000
80	+0.035	[+0.024, +0.046]	1.000
100	+0.035	[+0.025, +0.045]	1.000
150	+0.096	[+0.085, +0.110]	1.000
200	+0.121	[+0.108, +0.133]	1.000
250	+0.133	[+0.120, +0.145]	1.000
300	+0.150	[+0.138, +0.163]	1.000

nothing, because a respondent who selects the same option throughout is trivially caught by LongString and low IRV. On the *inconsistent* patterns—mixed and, especially, random—the full ensemble (green) sits well above the auxiliaries-only ensemble (red) across the whole corruption range: here the auxiliaries are partly blind, and ReReReRe rescues respondents they miss. The ensemble works not because every detector helps everywhere, but because the detectors fail on disjoint sets of respondents, and the Random Forest learns to route each case to whichever feature is informative for it.

4.8 Summary

Combining ReReReRe with four established auxiliary detectors inside a Random Forest yields a single careless-detection model whose pooled performance ($\text{MCC} = 0.66$, $\text{AUC} = 0.92$) scales from $\text{MCC} = 0.44$ on 30-item questionnaires to $\text{MCC} = 0.82$ on 300-item batteries under a deployable, label-free operating point. Ablation with bootstrap confidence intervals establishes that the contribution of ReReReRe is statistically positive at every questionnaire length and grows from +0.02 MCC on short instruments to +0.15 MCC on long ones. The mechanism is complementarity: ReReReRe rescues the inconsistent careless patterns (random, mixed) on which the auxiliary detectors are weak, while contributing little on the consistent patterns they already catch. What remains is the question the headline numbers quietly presupposed—how to set the operating point *without* the ground-truth labels that the cross-validation here had access to. That is the deployment problem, and it is the subject of Chapter 5.

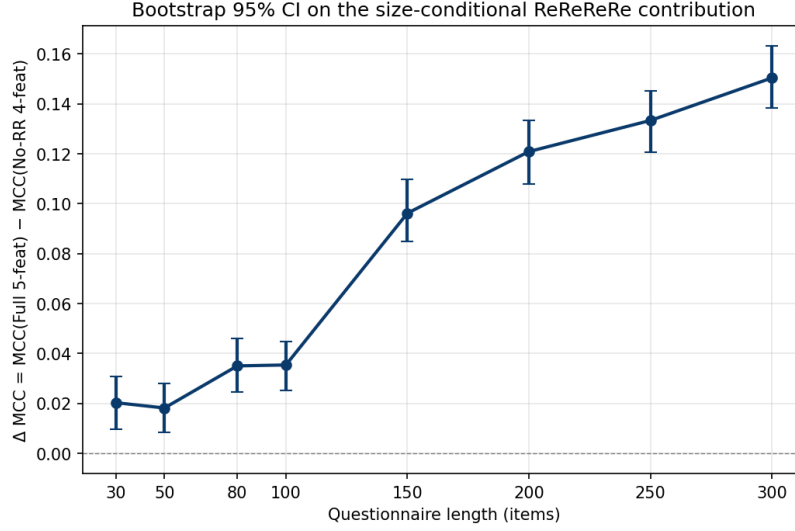


Figure 4.2: Bootstrap 95% confidence interval on the size-conditional contribution of ReReReRe ($\Delta\text{MCC} = \text{full} - \text{auxiliaries only}$). The interval excludes zero at every questionnaire length; the contribution grows from +0.020 at 30 items to +0.150 at 300 items.

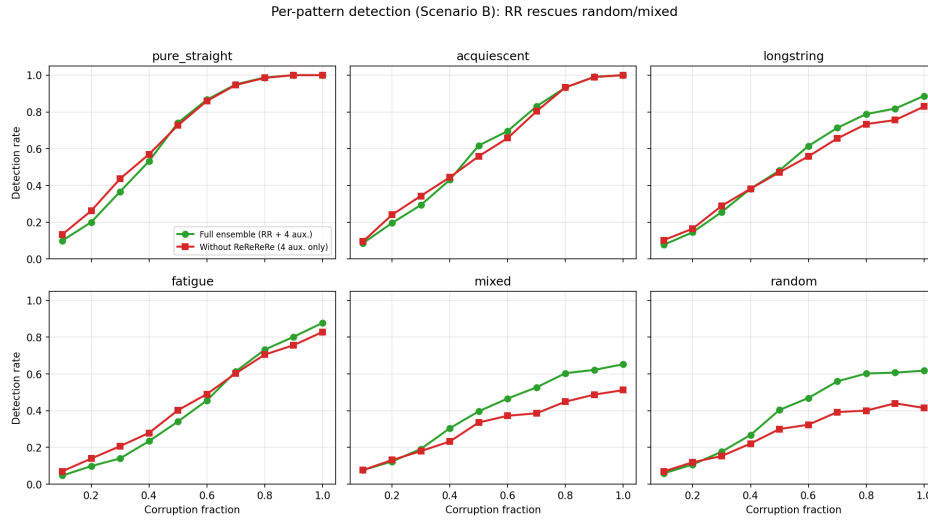


Figure 4.3: Per-pattern detection: full ensemble (green) versus auxiliaries only (red), by corruption fraction. The two coincide on consistent patterns (pure straightlining, acquiescence, longstring, fatigue), which the auxiliaries already detect; they diverge on the inconsistent patterns (mixed, random), where ReReReRe supplies the signal the auxiliaries lack.

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